Neelima Palleboina

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SUMMARY

Data Scientist with expertise in developing sophisticated analytics solutions and machine learning models that drive organizational optimization. Experienced in eliciting requirements, performing validation, and documenting analysis across complex datasets. Skilled at applying statistical methods to identify opportunities and present actionable insights that bridge the gap between technical analysis and business decision-making. Demonstrates intellectual curiosity and meticulous attention to detail while developing and implementing data-driven solutions across multiple domains.

EDUCATION

George Mason University **Virginia, USA**

Master's in Data Analytics Engineering, ***GPA:*** *3.94* ***August '22 - May '24***

COURSEWORK: Financial Analytics | Operational Research | Machine Learning & Data Mining | Data Visualization | statistics | Big Data Technologies | Business Intelligence

SKILLS

Programming Languages:Python, R, SQL, Java.

Cloud & Data Technologies:Azure, ETL, Google Cloud Platform (GCP), Big Query, AWS, Cloud Storage, Apache Spark, Hadoop, Databricks, S3, Regression, Git source control.

Machine Learning : A/B Testing, Statistical Analysis, Hypothesis Testing, Experimentation, Regression Models, User Segmentation, Cohort Analysis, Time Series Analysis, Funnel Optimization, NLP, Generative AI, HDFS, Generative Adversarial Network (GAN), Gen AI, LLM, Databricks

Tools :Tableau, Power BI, MS Excel(Data Analysis ToolPak, Power Query, Pivot Tables, VLOOKUP, HLOOKUP, INDEX MATCH),JIRA, Confluence.

WORK EXPERIENCE

The Parkside Group **August'24 – November'24**

Data Analyst- Contract

* Built comprehensive analytics experiments and reports using SQL and Python to extract insights from large datasets of election campaign information, guiding strategic decision-making across multiple teams.
* Applied statistical techniques and A/B testing to analyze funnel optimization, user segmentation, and cohort analyses, providing data-driven recommendations to improve campaign performance metrics by 15%.
* Created executive-facing dashboards in Power BI and Tableau to visualize time series analyses and regression models, reducing reporting time by 65% while effectively communicating marketplace dynamics and long-term trends to business leaders.

CATSR **January '24 – May '24**

Data Scientist - Contract

* Engineered and deployed a Generative AI (GAN) to produce over 2 million synthetic flight tracks, improving collision risk modeling accuracy by using Google Collab, Hopper Cluster for computing.
* Developed an ML pipeline with a Zurich runway dataset with 500 unique flight tracks, using Recurrent Neural Networks (RNN),Gated Recurrent Units (GRU) to enhance sequence prediction accuracy by 80%.
* Achieved a simulation accuracy of 92% for flight track generation, focused on system integration and iterative model refinements to elevate prediction precision and operational effectiveness by 75%.

Saven Technologies **June'18 – August '22**

Sr.Data Analyst

* Built full-cycle analytics experiments using SQL, Python and statistical tools to analyze complex ERP data, uncovering patterns that guided business decisions and improved revenue metrics by 78%.
* Applied statistical techniques including regression models, cohort analyses, and time series analyses to transform customer data, validating findings through hypothesis testing and experimentation to drive measurable business improvements.
* Created executive-facing dashboards with Tableau and visualization tools to track marketplace dynamics and user behaviors, collaborating with engineering teams to implement and validate data logging while maintaining data integrity across systems.

PROJECTS

Document Classification System with LLMs | *GenAI Implementation*

* Engineered a document classification solution using fine-tuned LLMs to automatically categorize and extract key information from technical documentation, improving processing efficiency by 40%.
* Implemented Supervised Fine-Tuning (SFT) techniques on a base model with a custom dataset of 5,000 annotated documents, achieving 88% classification accuracy across 12 document types
* Created a comprehensive validation framework with human-in-the-loop feedback mechanisms to continually improve model performance, documenting both technical metrics and business impact for senior stakeholders.

E-Commerce Analysis | *Machine Learning*

* Led a project to cut carbon emissions in the automotive industry, reducing environmental impact by 30%.
* Utilized a logistic regression model to predict car sales patterns, reducing surplus manufacturing by 25%, and provided insights on regional car model preferences, enhancing data acquisition efficiency by 20%.
* Delivered insights on regional car model preferences, contributing to a 20% data acquisition efficiency.

VOLUNTEERING

WoMen Of Connections Ministry **November '24 – Present**

Data Research Analyst

* Designed and executed experiments to test donation pattern hypotheses, implementing A/B testing methodologies that optimized resource distribution efficiency by 35% for the Token of Love project
* Developed comprehensive documentation and tracking systems for 8 data collection processes, including automated web scraping workflows using Python that improved information gathering by 60%
* Created and maintained Power BI dashboards connecting data sources, enabling stakeholders to track key metrics and make data-driven decisions that improved resource allocation effectiveness by 42%

CERTIFICATIONS

Machine Learning by Data Camp |SQL Advanced by Data Camp | Tableau Desktop | Power BI by Data Camp | Python for Data Science and Machine Learning by Google | Business Intelligence and Analytics Professional | Organizational Behaviour and Strategic Planning